



The Insolation — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Chapter 36_The_Insolation

Topic Index

1. Atmosphere — Sun se Indirect Heat
2. Height ke Saath Temperature Decrease
3. Albedo — Reflecting Power
4. Tropical Cyclones — Sea se Distance
5. Ocean Size aur Annual Temperature Range
6. Isotherms — January Deviation
7. Highest Temperature — 20°N Latitude
8. Clear-Sky Nights — Cooler
9. Infrared Waves — Water Vapour Absorption
10. Urban Heat Island

Atmosphere — Sun se Indirect Heat

- Atmosphere Sun se mostly **indirect heat** leta hai, Earth surface se **direct** (A TRUE) — short-wave solar energy ka long-wave terrestrial energy mein conversion **Earth surface pe hota hai** (R TRUE, sahi explanation) — CO₂/greenhouse gases long-wave radiation absorb karte hain, isse global warming hoti hai. [Q1]

Height ke Saath Temperature Decrease

- Height badhne se temperature ghatne ka reason: **(1) atmosphere mainly Earth surface se upward heat hoti hai, (3) upper atmosphere mein air density kam hoti hai** — **(2) upper atmosphere mein zyada moisture galat hai** (actually kam hoti hai). [Q2]

Albedo — Reflecting Power

- **Fresh snow-covered land** sabse zyada sunlight reflect karti hai (albedo ~75-95%) — sand desert (20-30%), paddy cropland (3-15%), prairie land (10-20%) se zyada. Earth ka average albedo ~30%. [Q3]
- **Albedo = Reflecting power** se associated hai — surface se reflect hone wale light ka fraction. Frozen ice ka albedo high, green areas (forests/plains) ka relatively low. [Q5]

Tropical Cyclones — Sea se Distance

- Sea se door jaane par tropical cyclones ki force **ghatti hai** (Statement 1 TRUE) — cyclones energy **latent heat (moisture condensation)** se leti hain jo ocean pe collect hoti hai; sea se door moisture supply/energy ghatti hai (Statement 2 TRUE, sahi explanation). [Q4]

Ocean Size aur Annual Temperature Range

- Ocean ka size jitna bada, annual temperature variation utna kam hota hai — **Atlantic Ocean** mein Pacific se zyada annual range hoti hai (Pacific bada hai) — Statement 1 (Pacific mein zyada) galat; **Northern Hemisphere** mein **Southern** se zyada annual range hoti hai (zyada land, Statement 2 TRUE). [Q6]

Isotherms — January Deviation

- January mein isotherms **ocean pe north ki taraf, continent pe south ki taraf deviate** karte hain (Statement 1 TRUE, especially Northern Hemisphere mein pronounced kyunki zyada land hai) — North Atlantic mein **Gulf Stream aur North Atlantic Drift WARM currents hain** (cold NAHI), isliye N.Atlantic ko warmer banate hain, isotherms north ki taraf bend karte hain (Statement 2 galat, 'cold currents' likha tha). [Q7]
- Statement I: January mein N.Hemisphere mein isotherms land ke upar equatorward, ocean ke upar poleward mudte hain — **TRUE**; Statement II: same latitude pe January mein ocean ki air land se **warmer** hoti hai — **TRUE** (land-sea thermal contrast hi isotherm bending ka physical reason hai, isliye II sahi explains I). [Q8]

⚠ EXAM TRAP

- Gulf Stream aur North Atlantic Drift **WARM currents** hain, cold NAHI — is fact ko galat likha jaana common trap hai. [Q7]

Highest Temperature — 20°N Latitude

- Maximum temperature zone Equator pe NAHI hota — actually **~20°N latitude** ke around record hota hai (thermal equator geographic equator se thoda north mein shift hota hai). [Q9, Q10]

⚠ EXAM TRAP

- Highest temperature Equator pe NAHI, **~20°N** pe record hoti hai — thermal equator geographic equator se north mein hoti hai. [Q9, Q10]

Clear-Sky Nights — Cooler

- Clear-sky nights cloudy nights se zyada thandi hoti hain kyunki **radiation** (terrestrial radiation clear sky mein freely space mein escape ho jaati hai — cloudy night mein clouds heat ko trap karke wapas radiate karte hain, isliye cloudy nights warmer feel hoti hain). [Q11]

Infrared Waves — Water Vapour Absorption

- Infrared waves mainly **lower atmosphere mein concentrated water vapour se absorb** hoti hain (troposphere mein water vapour, ozone, other gases) — short-wave solar energy relatively easily atmosphere se guzar jaati hai. [Q12]

Urban Heat Island

- **Urban Heat Island** = city ka average surface/atmospheric temperature **nearby rural areas se significantly zyada** hona — concrete/asphalt surfaces, low vegetation, reduced evapotranspiration, vehicles/industries se extra heat iske main reasons hain. [Q13]

